

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 2: Literature Review and Theoretical Framework

by

TAEHONG KWON

Student number: 28576810

submitted in accordance with the requirements for the degree of

DOCTOR OF PHILOSOPHY

in the subject

COMPUTER SCIENCE

at the

UNIVERSITY OF SOUTH AFRICA

SUPERVISOR: Prof. Ernest Mnkandla

CO-SUPERVISOR: Prof. Donatien Koulla Moulla

CO-SUPERVISOR: Dr. David Sena Attipoe

July 2026

2 Literature Review and Theoretical Framework

The research problem in Chapter 1 is that transfer-graph reputation scores mix payment flow with trust. Objectives O1–O5 and research questions RQ1–RQ5 follow from that problem. EndorseRank and Adaptive Weighted PageRank (AWP) are compared on that problem by walking their graphs with PageRank and checking the rankings with rank correlation. Hyperlink–citation propagation, domain alignment and computational complexity are the framework. The remaining gap is taken up in Chapters 3 and 4.

2.1 Blockchain and DeFi foundations

A public blockchain stores transactions in hashed blocks that later blocks lock in place ^[5]. Once a consensus of participants has accepted a block, rewriting it is expensive, so counterparties can settle against the ledger without sharing a legal identity (*trust-minimized* settlement).

Bitcoin used that ledger for peer-to-peer electronic cash ^[5]. Ethereum kept the same append-only history but let an account hold code as well as a balance, so a transaction can call a function and change contract state ^[7]; those calls run on the Ethereum Virtual Machine (EVM) ^[36]. The programs stored on the ledger are smart contracts, and every honest node that replays a call obtains the same result ^[6], which is why collateral checks, liquidations and fee rules can live in the contract rather than with a central operator ^[1].

The contracts that do this work are collectively called Decentralized Finance (DeFi). They cover exchange, lending, derivatives and asset management; Werner et al.^[8] systematise the protocol classes and Schär^[1] describes the stack of layers they sit on. Because one protocol can call another, new products can be assembled from existing ones ^[37], and failures can travel the same way. Flash-loan attacks ^[38] and price-oracle manipulation ^[39] are documented cases. Wallet histories on these

chains are public and machine-readable, but they record actions, not legal identities ^[2].

2.1.1 Pseudonymity and the credit problem

Traditional credit scoring links a legal person to a repayment history. On a public blockchain the unit of analysis is an address (a wallet), which may belong to a person, a fund, a bot or a contract. One entity can control many addresses ^[11], and clustering heuristics recover that control only in part ^[40]. Lending markets therefore ask borrowers to lock more collateral than they borrow, and liquidators seize that collateral when health factors fall ^[41]. The arrangement is robust and capital-inefficient. On-chain reputation scores try to summarise behavioral reliability so that a protocol, or a researcher studying it, can reason about risk without off-chain KYC. Published proposals include prospect-theory credit models ^[13], machine-learning scores on lending data ^[14] and risk propagation over account networks ^[12].

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Ethereum mainnet is where DeFi first concentrated security and liquidity, but demand pushes up fees and limits throughput ^[42]. Layer-2 (L2) protocols try to keep that security while moving execution off the base chain, or compressing it, and posting commitments back to Ethereum ^[43]. Thibault et al.^[44] survey how rollups do this. Optimistic rollups such as Arbitrum treat a posted batch as correct unless someone challenges it during a dispute window ^[45].

Arbitrum One is one such EVM-compatible rollup. It is widely used for DeFi, including the GMX perpetual-swap venue ^[46] whose contracts are among the most-approved spenders in the observation window. Perpetual futures are a separate derivative class; Ackerer et al.^[47] analyse their funding-rate pricing. This research uses Arbitrum One because ERC-20 Approval and Transfer logs are in public BigQuery datasets and because fees are low enough that six months of activity is dense. Results should not be assumed to transfer automatically to other L2s or to Ethereum mainnet without re-validation, but the template (allowance graph versus transfer graph, proxy families, rank

correlation) applies on any chain that emits the same logs.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens: `transfer`, `approve`, `transferFrom`, `allowance`, and related events ^[18]. When an owner calls `approve(spender, amount)`, the token contract records that `spender` may transfer up to `amount` from the owner. The spender later calls `transferFrom` to move tokens. EIP-2612 permit allows allowance updates via off-chain signatures settled in a single transaction ^[19], improving user experience without changing the fundamental authorization semantics.

Figure 2.1 illustrates the lifecycle. An allowance is an *explicit authorization*, not a completed transfer. Owners may set allowances to routers, vaults, or trading contracts they trust to act within limits. Revocation (approving zero) and replacement (a new non-zero allowance) update the live authorization state. EndorseRank therefore uses the *latest* allowance per owner–spender pair as edge strength: older approvals are superseded rather than decayed continuously.

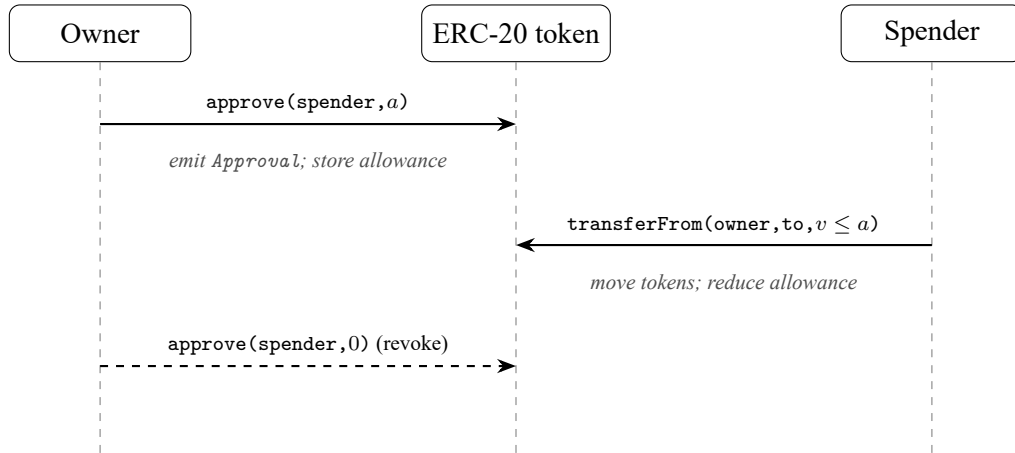


Figure 2.1: ERC-20 allowance lifecycle: approve (or permit), optional `transferFrom`, and revoke. EndorseRank edges reflect the latest live allowance snapshot per owner–spender pair (historical approvals superseded, not time-decayed).

From a measurement perspective, allowances are sparser than transfers: many transfers occur under a single approval, and many wallets never approve third-party spenders. This sparsity is both a

computational advantage and a semantic claim: EndorseRank measures who is authorised to spend, whereas a transfer graph measures who pays whom.

2.4 Graph-based on-chain reputation

PageRank and variants rank nodes by propagated influence through directed edges^[15]. Two recent lines of work apply this idea to blockchain transaction graphs and must be kept distinct. Do and Do^[4] apply PageRank and HodgeRank to Ethereum transaction graphs as a social-credit measure; their contribution is the framing of transfer connectivity as credit-relevant. Adaptive Weighted PageRank (AWP), proposed by Do, Do, and Nguyen^[3], is the transfer-graph baseline used in this thesis: it weights each transfer edge by value and by a logistic time-decay of edge age, and it validates the resulting scores against in-degree and in-value (count and sum of inbound transfers) as domain-intuition proxies for economic importance. The logistic decay formulation and the in-degree/in-value validation protocol reproduced in Chapter 3 are therefore attributed to the AWP paper^[3], not to Do and Do^[4]. Lin et al.^[12] show that transaction connectivity alone can miss risk-relevant structure unless risk signals propagate through the network, motivating refined graph semantics beyond vanilla PageRank; Wu et al.^[24] reach a similar conclusion for phishing detection, where network embeddings of transaction graphs outperform per-account features. Surveys of blockchain-based reputation systems^[25] and of smart-contract reputation designs^[26] document that most deployed systems still rely on explicit ratings or transfer volume as the trust signal; a survey of knowledge discovery in cryptocurrency transaction data^[48] likewise lists degree- and value-based centralities as the dominant wallet descriptors. Packin and Lev-Aretz^[2] emphasize auditability constraints under pseudonymity. EndorseRank stays within interpretable graph propagation of that kind, so it can be audited in the way they require.

Graph-based approaches are attractive because they (i) use only public data, (ii) produce continuous scores suitable for ranking, and (iii) admit complexity analysis in terms of edge counts. They are limited by Sybil attacks^[11], wash trading, and the gap between graph centrality and economic

default risk. PageRank scores are not treated as credit scores in the regulatory sense. The claim is that allowance-based and transfer-based rankings can be compared rigorously on shared proxies.

2.5 PageRank formalism

Let $G = (V, E)$ be a directed graph with $|V| = n$ nodes. Write $w_{ij} \geq 0$ for the weight of edge $i \rightarrow j$ (influence from i toward j in the endorsement or transfer sense used here). Let W be the weighted adjacency matrix and define the out-strength $d_i^{\text{out}} = \sum_j w_{ij}$. The row-stochastic transition matrix P is

$$P_{ij} = \begin{cases} w_{ij}/d_i^{\text{out}} & \text{if } d_i^{\text{out}} > 0, \\ 1/n & \text{if } d_i^{\text{out}} = 0 \end{cases} \quad (2.1)$$

(dangling nodes distribute mass uniformly). The Google matrix with damping $d \in (0, 1)$ is

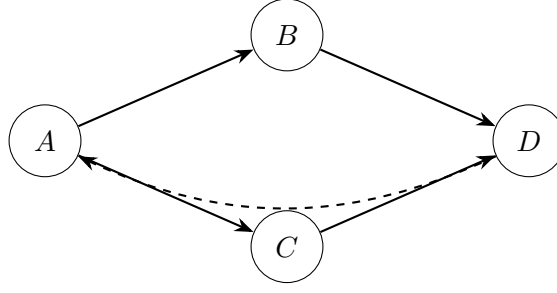
$$G_{\text{PR}} = d P^\top + \frac{1-d}{n} \mathbf{1}\mathbf{1}^\top, \quad (2.2)$$

and the PageRank vector π satisfies $\pi = G_{\text{PR}} \pi$ with $\pi \geq 0$ and $\sum_i \pi_i = 1$ ^{[15],[23]}. Equivalently,

$$\pi_j = d \sum_{i:(i \rightarrow j) \in E} \pi_i \frac{w_{ij}}{d_i^{\text{out}}} + \frac{1-d}{n}. \quad (2.3)$$

2.5.1 Random-surfer interpretation

Equation (2.3) admits a probabilistic reading: a surfer follows outgoing edges with probability d and teleports uniformly with probability $1 - d$ (Figure 2.2). The stationary distribution ranks nodes by long-run visit frequency. Damping prevents sinks from trapping all mass and guarantees primitivity of G_{PR} under mild conditions, so the Perron vector is unique ^[23].



solid: follow edge (prob. d); dashed: teleport (prob. $1 - d$)

Figure 2.2: Random-surfer intuition for PageRank. Solid arrows are graph transitions; dashed arrows represent teleportation that mixes the chain.

2.5.2 Power iteration and complexity

In practice π is computed by power iteration:

$$\pi^{(t+1)} = G_{\text{PR}} \pi^{(t)}, \quad (2.4)$$

until $\|\pi^{(t+1)} - \pi^{(t)}\|_1 < \varepsilon$ or a maximum iteration count is reached. Each iteration costs $O(|E|)$ arithmetic operations for sparse graphs [49]. Methods that induce fewer edges, such as latest-allowance snapshots in place of full transfer histories, therefore reduce per-iteration cost and often memory. This complexity observation grounds hypothesis H3.

Default parameters in this research use damping $d = 0.85$ following common practice [15], with convergence tolerance $\varepsilon = 10^{-8}$ and a maximum of 300 iterations as study-specific solver settings.¹ Robustness checks vary $d \in \{0.75, 0.85, 0.95\}$.

2.6 Reputation systems lineage

PageRank is one of a family of spectral and propagation-based reputation mechanisms. Kleinberg’s HITS algorithm ranks pages as hubs and as authorities [50]. EigenTrust takes the same idea into peer-

¹Tolerance and iteration caps are implementation choices for reproducible timing; they are not claimed as part of the classical PageRank formulation.

to-peer networks and normalises local trust so that a malicious node cannot inflate its own score at will ^[16]. TrustRank then starts from a seed set of known-good nodes and pushes that trust outward to demote web spam ^[17]. In each case reputation is relational: it is hard to forge without costly endorsements from nodes that are already reputable.

On-chain, Sybil attacks remain central ^[11]. Creating wallets is cheap; creating wallets that receive allowances or transfers from diverse counterparties is costlier. Sybil-adjusted stability proxies in this research (inbound counterparty ratio, tenure, active months) partially capture diversity of interaction but do not eliminate adversarial manipulation. Chapter 5 sketches a cost model for allowance-graph Sybils.

Machine-learning credit models on DeFi features ^[14] can outperform simple graph scores on labeled defaults when labels exist, but they trade interpretability for accuracy. EndorseRank and AWP prioritize transparent propagation rules suitable for audit ^[2].

2.7 ERC-20 allowances as endorsement

ERC-20 approve and EIP-2612 permit record explicit spending authorization ^{[18],[19]}. EndorseRank treats the latest allowance per owner–spender pair as edge strength on a directed endorsement graph and runs PageRank on that snapshot, eliminating historical time-decay because newer approvals override older ones.

Formally, for each token and owner–spender pair, let $a_{o \rightarrow s}$ be the decimal-adjusted latest allowance. Edges are aggregated across tokens by summing allowances into a single weight $w_{o \rightarrow s}$. The resulting graph $G_{\text{approve}} = (V, E_{\text{approve}}, w)$ is typically much sparser than the transfer graph over the same wallets and window. PageRank on G_{approve} yields EndorseRank scores s_{ER} .

The endorsement interpretation is intentional but imperfect. Users approve routers for convenience, not always as a social judgment of the router’s “trustworthiness.” Contract addresses and EOAs coexist. Nevertheless, allowances are a clearer authorization signal than passive receipt of transfers, which may reflect market making, wash trading ^[29], airdrop farming ^[32], or routing without any

endorsement intent.

2.8 What this research does not claim

This research does not claim that EndorseRank predicts trading profit, loss avoidance or liquidation. The research proposal planned to validate both methods against GMX V2 trading-success proxies and to bound that validation with an outcome-native reference (a PageRank on a star graph built from realized profit and loss). That reference is circular by construction: it is computed from the same PnL stream that defines the trading-success proxies, so its near-perfect agreement with the realized-gain proxy is the correlation of a variable with a monotone function of itself, not evidence about reputation. For the same reason, weak agreement between a social method and those proxies cannot be read as failure. The reference, the associated supplementary research question and claim, and the trading-success, inverse-risk and liquidation families are therefore withdrawn from the primary comparison; their matrices are archived in Appendix 8.4. The remaining external check is a temporal holdout of future new owner–spender approvals (Chapters 3 and 4), which is out of window but of the same construct as the score.

2.9 Rank correlation theory

Reputation scores and validation proxies are continuous variables without a universal classification threshold. Following the AWP validation design^[3] and classical psychometrics^[22], this research evaluates monotonic and pairwise rank agreement, and it quantifies the sampling uncertainty of every reported coefficient with bootstrap confidence intervals^[34].

Spearman’s rank correlation^[20] is Pearson correlation on ranks. For paired ranks (R_i, S_i) of n items,

$$\rho = 1 - \frac{6 \sum_{i=1}^n (R_i - S_i)^2}{n(n^2 - 1)} \quad (2.5)$$

in the absence of ties (with tie-aware variants used in software). Spearman’s ρ captures monotonic association.

Kendall’s τ ^[21] counts concordant and discordant pairs. For all pairs (i, j) with $i < j$,

$$\tau = \frac{2(C - D)}{n(n - 1)}, \quad (2.6)$$

where C (D) is the number of concordant (discordant) pairs, with tie corrections in the τ -b variant. Kendall’s τ is often more interpretable for pairwise ordering agreement and is the primary family-level summary in Chapter 4. Values are compared relatively across methods on the same cohort rather than against an uncited absolute threshold.

2.10 Theoretical framework

This study uses hyperlink–citation propagation, construct (domain) alignment, and computational complexity as the bases for comparing EndorseRank and AWP across reputation structure (H1), social-method alignment (H2), and computational efficiency (H3).

- Hyperlink–citation semantics: The original PageRank algorithm ^[15] treats inbound hyperlinks as citations. Do and Do ^[4] apply this logic to Ethereum transfers; AWP ^[3] uses time-decayed transfer edge weights. Figure 3.2 (Chapter 3) plots the logistic decay

$$\sigma(\Delta t) = \frac{1}{1 + \exp(k(\Delta t - t_0))} \quad (2.7)$$

with $k = 0.01$ and $t_0 = 180$ days as used for AWP in this study. EndorseRank extends citation-based propagation to ERC-20 approve/permit edges because allowances are explicit, on-chain delegations of spending authority ^{[18],[19]}. Unlike transfers, which may reflect trading or routing without trust intent, an allowance records explicit authorization that remains in force until revoked or replaced. Graph-based risk models ^[12] further show that network structure carries information beyond raw activity counts, which supports a directed endorsement graph in addition to transfer volume.

- Domain alignment framework: In measurement theory, construct alignment assesses how

well an indicator tracks an intended construct ^[22]; Messick^[33] extends this to a unified judgment about what a score means and what its use implies. For rank-based reputation scoring, Spearman’s ρ and Kendall’s τ evaluate monotonic and pairwise ordering agreement without assuming a classification threshold, as in the AWP validation protocol^[3]. Contemporaneous transfer and allowance statistics check graph coherence; a temporal holdout of future new approvals is the out-of-window check.

- Computational complexity: PageRank’s per-iteration cost is $O(|E|)$ ^[15]. Allowance graphs are sparser than transfer graphs (fewer edges $|E|$), so per-iteration cost and memory are lower and, in practice, fewer iterations are often needed; this is the basis for H3. Because the solver is identical for both methods, any runtime difference is an expected consequence of edge sparsity, not an algorithmic contribution.

Figure 2.3 presents the conceptual model mapping the primary EndorseRank–AWP comparison to H1–H3 and RQ1–RQ4; RQ5 sits on top of it as the question of whether the two constructs combine.

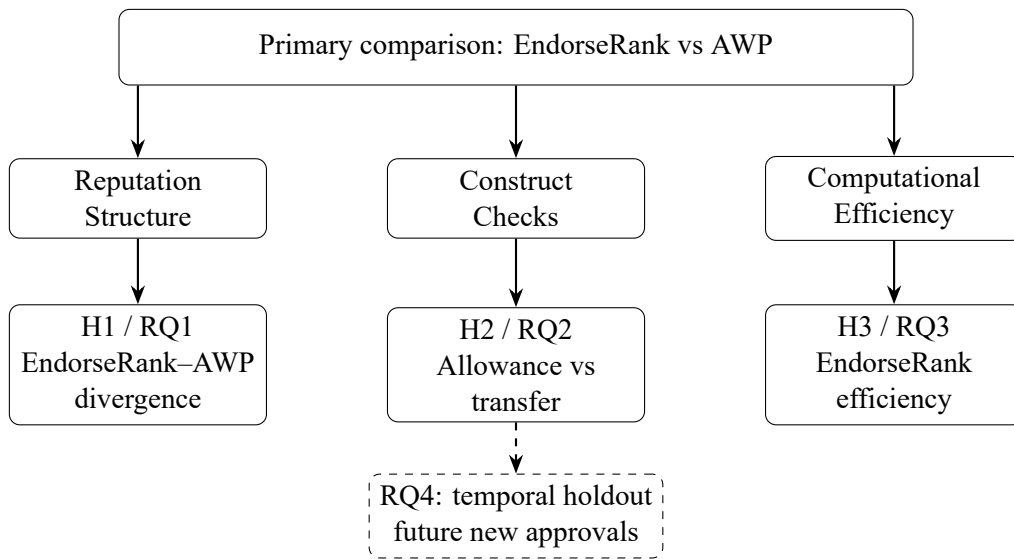


Figure 2.3: Conceptual model: primary EndorseRank vs. AWP comparison mapped to reputation structure (H1/RQ1), construct checks (H2/RQ2), computational efficiency (H3/RQ3), and the temporal holdout of future new approvals (RQ4).

H1 and H3 concern EndorseRank and AWP only; H2/RQ2 compares contemporaneous allowance

and transfer checks; RQ4 is a time-split test of future new owner–spender approvals on the spender cohort, not a ranking of trading outcomes.

2.11 Terminology

The definitions in Chapter 1 establish formal constructs. The following terms are used when interpreting results:

- Primary methods: EndorseRank (allowance graph) and AWP (transfer graph). Both are social PageRank methods on wallet-to-wallet edges; intended as deployable reputation signals.
- Primary graph construct: The graph a method is *designed* to rank (e.g., G_{approve} for EndorseRank). Distinct from checks applied after scoring.
- Contemporaneous check: Local transfer or allowance statistics computed in the same window as the score ^[22].
- Temporal holdout: Scores frozen at t_1 ; labels from t_2 new owner–spender approvals on inbound spenders.
- Matched wallet cohort: The primary same-window sample of $n = 5,521$ wallets with non-zero endorsement- and transfer-graph connectivity.
- Spender holdout cohort: $n = 1,335$ t_1 inbound allowance recipients used for RQ4.
- Expanded wallet pool: A larger address set ($N = 99,080$) used for the node-count sensitivity of runtime beyond the matched cohort.

2.12 Domain alignment metrics

Following the AWP validation protocol^[3] and Cronbach and Meehl^[22], construct alignment for rank-based scores uses Spearman’s ρ and Kendall’s τ between reputation scores and check rankings. This research reports contemporaneous transfer and allowance checks and a temporal holdout of future new approvals; see Chapter 3 for operational definitions.

The literature reviewed here supports three design choices: (i) PageRank as an interpretable propagation backbone; (ii) allowances as endorsement edges distinct from transfers; and (iii) rank correlation plus a time split as the check design. These choices motivate the remaining sections of this chapter and, later, the operational pipeline in Chapter 3.

2.13 Worked example: PageRank on a four-node endorsement graph

Consider four wallets $\{A, B, C, D\}$ with latest-allowance edges

$$A \rightarrow B (2), \quad A \rightarrow C (1), \quad B \rightarrow D (3), \quad C \rightarrow D (1),$$

and damping $d = 0.85$. Out-strengths are $d_A^{\text{out}} = 3$, $d_B^{\text{out}} = 3$, $d_C^{\text{out}} = 1$, $d_D^{\text{out}} = 0$ (dangling). After one power-iteration step from the uniform vector $\pi^{(0)} = \mathbf{1}/4$, mass flows preferentially toward D because both B and C endorse D , while A splits mass between B and C . Subsequent iterations mix in teleportation mass $(1 - d)/4 = 0.0375$ at every node. The stationary ranking places D highest, then B and C , then A . The example shows that *receiving* endorsements from nodes that themselves receive endorsements raises score, which is exactly the citation logic of PageRank ^{[15],[23]}.

If the same wallets are connected by many small transfers instead of few allowances, the transition matrix changes and the ranking can reorder even though the node set is identical. That is the microcosm of H1: EndorseRank and AWP need not agree.

2.14 DeFi risk taxonomy and where reputation fits

Werner et al.^[8] organise DeFi risk by protocol layer and Zhou et al.^[27] by the attack surfaces seen in real incidents. A contract bug or an upgrade key is a smart-contract risk ^[51]. A stale or manipulated price is an oracle risk ^[39]. Volatility and liquidation cascades, including stress in stablecoins and funding markets ^[53], are market risks ^[52]; the March 2020 crash showed how those layers interact ^[9]. Token-weighted voting can be captured, and that is a governance risk ^[54]. This research is about

a narrower layer, *counterparty* or *behavioral* risk: whether a wallet’s past interactions suggest that it manages leveraged exposure reliably.

On-chain reputation scores address only the behavioral layer, and only partially. They do not replace collateral, audits, or oracle design. Their role is to summarize relational structure (who authorizes whom; who pays whom) into an ordinal signal that can be audited and recomputed [2]. EndorseRank targets the authorization facet of behavioral risk; AWP targets the payment-flow facet. Neither claims to predict exploits or governance attacks.

Artificial activity further complicates behavioral measurement. Wash trading inflates volume on decentralized exchanges [28] and, at scale, on unregulated centralized venues [29]; front-running and other forms of extractable value generate transfers that reflect ordering games rather than relationships [55], and Qin et al. [30] quantify how much of that value is realised on Ethereum. Scam tokens on Uniswap [31] and cross-venue arbitrage flows [56] likewise inflate transfer activity without durable counterparty relationships. Transfer graphs are especially exposed to artificial volume; allowance graphs are exposed to approval farming. Time-split checks and Sybil-oriented diagnostics are partial mitigations, not complete defenses.

2.15 Related measurement traditions

Beyond blockchain, reputation systems have been studied in e-commerce [57], peer-to-peer networks [16], and web spam [17]. Common lessons include: (i) reputation is relational; (ii) cheap identities enable Sybil attacks [11]; (iii) seed sets and normalization limit inflation; and (iv) evaluation requires external criteria, not only internal consistency [22]. EndorseRank inherits (i)–(iii) via PageRank damping and endorsement costs, and addresses (iv) via contemporaneous construct checks measured with Spearman’s ρ [20] and Kendall’s τ [21], and via a temporal holdout.

Personalized and topic-sensitive PageRank variants [58],[59] bias the teleportation vector toward a chosen set of nodes, which is how TrustRank turns a small seed set into a global score [17]. Two families of work extend PageRank to graphs with more than one edge type. In Multiplex PageRank, the

centrality of a node on one layer conditions the walk on another ^[64]; in the multilayer formulation of De Domenico and colleagues, a single walker moves within and between layers of one interconnected network ^[65]. Both treat the layers as views of the same node set, which is the situation here: allowance edges and transfer edges are two relations over one address space. EndorseRank and AWP use uniform teleportation and one layer each, for comparability with the baseline ^[3]. The coupled operator of Chapter 3 takes the simplest multilayer form, a per-node convex mixture of the two transition rows with one weight, and the seeded ablation applies the personalised-teleportation idea with the allowance score as the seed vector.

2.16 Accounts, gas, and why logs are the measurement surface

Ethereum-style accounts are either externally owned accounts (EOAs), controlled by private keys, or contract accounts, controlled by code ^[60]. Every state-changing interaction consumes gas and emits logs when contracts are written to do so. Reputation research that aims for auditability therefore privileges *logs* over off-chain indexes: logs are part of the canonical chain history and can be re-derived by any full node or public warehouse ^[61].

Gas costs shape the observed graphs. On L2s, lower fees increase transfer frequency and approval churn, densifying G_{transfer} more than G_{approve} because payments are routine while approvals are intermittent setup actions. This economic asymmetry is one reason EndorseRank graphs are expected to remain sparse relative to transfer graphs (examined in Chapter 4).

2.17 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) are a well-known user-experience hazard: a compromised spender can drain tokens until the owner revokes. Security analyses of honeypot contracts ^[62] and of phishing accounts on Ethereum ^[24] treat allowances as capabilities, not as neutral metadata. EndorseRank interprets inbound allowances as endorsement of a spender’s capability, which is precisely why the construct is semantically distinct from inbound transfers. A wallet may receive

many transfers without ever being trusted to spend on others' behalf.

From a measurement standpoint, unlimited approvals create heavy-tailed edge weights. Decimal adjustment and optional winsorization can moderate extremes; this research sums decimal-adjusted latest allowances and relies on PageRank's normalization by out-strength so that a single huge approval does not automatically dominate unless the owner has few outgoing endorsements.

2.18 AWP in depth: what the transfer-graph baseline measures

Do and Do^[4] apply PageRank and HodgeRank to Ethereum transfers as a social-credit signal. Adaptive Weighted PageRank (AWP), introduced by Do, Do, and Nguyen^[3], uses transfer edges whose weights combine value and recency through a logistic decay, and validates the ranking against inbound degree and inbound value, proxies for "how much economic attention a wallet receives." Both the decay formulation (Equation (2.7)) and the in-degree/in-value proxies used in Chapters 3 and 4 come from the AWP paper. The framework is attractive because it is fully on-chain, rank-based, and free of off-chain identity.

EndorseRank preserves the PageRank-and-rank-correlation template but replaces the edge semantics. Chapter 4 tests whether allowance edges produce a construct-distinct, allowance-aligned and cheaper score, and whether t1 scores predict t2 new approvals.

2.19 Comparison of graph reputation methods

Table 2.1 situates EndorseRank among related methods discussed in this chapter by edge semantics, temporal model, and validation design.

Table 2.1 shows that EndorseRank's novelty is the choice of latest-allowance endorsement edges as the primary social construct for DeFi wallets, not a new ranking algorithm. The coupled operator and its seeded ablation are the point at which this thesis does alter the operator, by bringing the two relations into one walk; the test of that change is out-of-window, against both constructs, under a rule fixed in advance.

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method	Edge semantics	Temporal model	Typical validation
PageRank [15]	(web) Hyperlinks	None / static snapshot	Retrieval quality
EigenTrust [16]	Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank [17]	Links from seed goods	Static / seeded	Spam demotion
PageRank/HodgeRank on Ethereum [4]	Token transfers	Static window	Not reproduced in this thesis
AWP [3]	Token transfers	Logistic decay	In-degree / in-value
RiskProp [12]	Risk-bearing links	Propagation of risk	Account risk labels
Multiplex / multilayer PageRank [64],[65]	Several relations over one node set	Static layers	Versatility of top-ranked nodes
EndorseRank (this study)	Latest allowances	Snapshot (override)	Allowance checks; t2 new approvals
C-PR / S-PR (this study)	Allowances mixed with, or seeding, transfers	Snapshot and logistic decay	Both holdout labels; pre-registered rule

2.20 Summary of the reviewed literature

Table 2.2 consolidates the sources that shape the design of this study. For each source it records the methodology, the strengths that this thesis borrows, the limitations that it tries to address, and the relation to the present work. The table is the evidence base for the research gap stated in Section 2.21.

Table 2.2: Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study.

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Page et al. (1998) [15]; Langville and Meyer (2006) [23]	Eigenvector ranking of a hyperlink graph; power iteration	Interpretable propagation; $O(E)$ per iteration; unique Perron vector under damping	Edge semantics fixed to hyperlinks; no notion of time	Solver and complexity argument reused unchanged for both compared methods

continued on next page

Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Kamvar et al. (2003) ^[16]	Trust aggregation in P2P networks; seeded propagation against spam	Normalisation and seed sets limit inflation; explicit adversary model	Requires explicit ratings or curated seeds; not on-chain	Motivates the Sybil-cost discussion and the uniform-teleportation baseline
Gyöngyi et al. (2004) ^[17]				
Do and Do (2023) ^[4]	PageRank and HodgeRank on Ethereum transaction graphs as social credit	Framing of transfer connectivity as credit-relevant; fully public data	Transfers conflate payment and trust; no allowance semantics	Prior work on the transfer graph; distinguished from AWP in this thesis
Do, Do, and Nguyen (2023), AWP ^[3]	Value- and recency-weighted transfer PageRank with logarithmic decay; rank correlation against in-degree/in-value	Rank-based validation without default labels; recency handled explicitly	Proxies are computed from the same transfer data that builds the graph (mechanical alignment); dense graph	Baseline method, decay formula, and proxy protocol reproduced exactly on the same cohort
Lin et al. (2025), RiskProp ^[12]	De-anonymisation score plus network propagation of account risk	Shows structure carries information beyond activity counts; labelled evaluation	Needs risk labels; targets fraud, not trust or authorisation	Motivates refined edge semantics; RiskProp-style propagation kept as an archived extended baseline
Wu et al. (2022) ^[24]	Network embedding of Ethereum transaction graphs for phishing detection	Graph features outperform per-account features; large labelled set	Supervised; labels are phishing, not credit-worthiness	Supports the premise that graph position is informative; not a competitor method
Hassija et al. (2020) ^[13] ; Kotb (2023) ^[14]	Prospect-theory and machine-learning credit scoring on blockchain data	Directly target lending decisions	Require labels or off-chain features; limited interpretability	Contrast class: EndorseRank stays within auditable graph propagation

continued on next page

Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Bellini et al. (2020) ^[25] ; Almasoud et al. (2020) ^[26]	Surveys of blockchain-based trust and reputation systems	Map the design space; identify explicit-rating and volume-based signals as dominant	Do not evaluate authorisation edges; no L2 measurement	Establish that latest-allowance edges are absent from the surveyed designs
Victor (2020) ^[40] ; Victor and Weintraud (2021) ^[28] ; Cong et al. (2023) ^[29]	Address clustering heuristics; wash-trading detection on DEXs and exchanges	Quantify how much on-chain activity is artificial or multi-address	Do not propose a reputation score	Explain why transfer-based signals are exposed to inflation; motivate the allowance construct
Cronbach and Meehl (1955) ^[22] ; Messick (1995) ^[33]	Construct validity in psychometrics	Separate what a score is designed to measure from external criteria	Not specific	Frame the contemporaneous checks as construct checks and the temporal holdout as external evidence
Efron (1979) ^[34] ; DiCiccio and Efron (1996) ^[35]	Bootstrap resampling and bootstrap confidence intervals	Distribution-free uncertainty for rank statistics; paired contrasts	Assumes exchangeable units	Provides the 95% intervals and $\Delta\tau$ contrasts reported in Chapter 4

2.21 Research gap

Five gaps follow from Table 2.2 and Section 2.15. First, every on-chain reputation method reviewed (PageRank/HodgeRank ^[4], AWP ^[3], RiskProp ^[12], and the systems catalogued in the surveys ^[25]) builds its graph from transfers or explicit ratings; none uses the ERC-20 allowance, the only on-chain event that records a deliberate delegation of spending authority. Whether a latest-allowance endorsement graph produces a different and useful ordering is therefore untested (addressed by O1 and RQ1). Second, the validation protocol of the transfer-graph baseline compares scores with

in-degree and in-value computed from the same transfer data that builds the graph, so agreement is partly mechanical; no study reports the same protocol on an allowance construct alongside its transfer counterpart with confidence intervals and paired difference tests on one cohort (O2 and RQ2). Third, reported runtimes for blockchain PageRank are rarely accompanied by edge counts, memory, and iteration counts under a fixed protocol, which makes efficiency claims hard to attribute to sparsity versus implementation (O3 and RQ3). Fourth, a methodological gap concerns time: same-window graph statistics are routinely presented as validation even though they cannot show that a score anticipates future behaviour; a temporal holdout with a future-tense label for each construct, and the raw degree at the freeze date as the baseline, is the out-of-window check adopted here (O4 and RQ4). Fifth, multilayer ranking has been developed on social, transport and citation networks ^{[64],[65]} but has not been applied to two on-chain relations over one address set, and no study reports whether mixing an authorisation layer with a payment layer improves on either layer alone under a rule fixed before the result is seen (O5 and RQ5).

2.22 Open problems motivating this research

Several open problems in on-chain reputation remain unresolved in the literature and motivate the present study:

1. Edge semantics: Which on-chain relations best encode trust or authorization under pseudonymity ^[2]?
2. Validation without defaults: How can methods be compared when loan-default labels are scarce ^[4]?
3. Computational cost: Can reputation scores refresh at interactive latencies as graphs grow ^[23]?
4. Time-split checks: How can evaluation avoid mistaking same-window graph statistics for out-of-window evidence ^[22]?
5. Adversarial robustness: What is the cost of forging reputation under cheap identities ^[11]?

EndorseRank addresses (1)–(4) with allowance edges, contemporaneous checks, runtime bench-

marks, and a temporal holdout, and analyzes (5) at a model level in Chapter 5. Full empirical adversarial evaluation remains future work.

Chapter 3 turns the gap statement into an operational design: it fixes the data window and cohort, specifies how both graphs are constructed and solved with one shared PageRank implementation, defines the proxy families and the temporal holdout, and sets the timing and bootstrap protocols whose outputs Chapter 4 reports.